

Time: 30 minutes

Max marks: 10

Instructions:

- Do not plagiarize. Do not assist your classmates in plagiarism.
- Show your full solution for the questions to get full credit.
- Attempt all questions that you can.
- In the unlikely case a question is not clear, discuss it with an invigilating TA. Please ensure that you clearly include any assumptions you make, even after clarification from the invigilator.

1. (2 points) Argue for or against the following two statements. Explain your stand.
 - (a) (1 point) A deep CNN based image classification model that contains fully connected layers can take an input image of arbitrary size.
 - (b) (1 point) A fully convolutional network that does not contain any fully connected layers can take an input image of arbitrary size.

Solution:

- (a) (1 point) A CNN based model that contains fully connected layers *can not* accept arbitrary sized input images as the feature map in the layer preceding the fully connected layers has to contain the exact number of elements as the weights of the fully connected layer. A larger or smaller image will lead to a mismatch in the number of inputs and weights of the fully connected layer, leading to an error.
- (b) (1 point) Yes, a fully convolutional network model can process an arbitrary sized image as input and will generate feature maps that are of corresponding size.

2. (2 points)
 - (a) (1 point) How does DeepLabV3+ incorporate a *lightweight* decoder in its architecture?
 - (b) (1 point) What are the other alternatives considered and why were they discarded?

Solution:

- (a) (1 point) The lightweight decoder in the DeepLabV3+ architecture combines the upsampled feature map from the output of the encoder with intermediate features from the encoder, followed by upsampling to the input image's size.
- (b) (1 point) The other alternatives were a single-shot upsampling (like in Fully Convolutional Network (FCN)) and a full-blown U-Net style decoder where feature maps from multiple layers of the encoder were combined with the decoder layers' feature maps. The former (FCN) has very few learnable parameters and results in poorer performance, whereas the latter (U-Net) has too many learnable parameters. The lightweight decoder in DeepLabV3+ was a reasonable middle ground.

3. (2 points) Explain what were *skip connections* in the following architectures and what did they help achieve?
 - (a) (1 point) ResNet architecture for image classification.
 - (b) (1 point) Fully Convolutional Networks for semantic segmentation.

Solution:

(a) (1 point) (0.5) Architecture (0.5) Achievement

1. The skip connections in ResNet were effective in training deeper network, In ResNet, skip (residual) connections add the input x directly to the output of convolutional layers $F(x)$, allowing the network to learn the residual mapping $H(x) = F(x) + x$ and improving gradient flow in deep networks.
2. Steady improvement in classification accuracy. Largely resolved the vanishing/exploding gradient problem, allowing networks to scale up to 152 layers. The network can learn an identity mapping if a layer is not useful ($F(x) = 0$), enabling effective feature reuse.

(b) (1 point) (0.5) Architecture (0.5) Achievement

1. In Fully Convolutional Networks (FCNs), skip connections fuse upsampled deep semantic feature maps with earlier convolutional feature maps containing fine spatial details, typically combining them as $Output = Upsample(Deep\ Features) + Shallow\ Features$.
2. Improved the quality of segmentation masks by restoring fine spatial details, leading to better localization and higher segmentation performance (e.g., improved mIoU).

4. (4 points) Answer the following questions. Define all terms that you introduce for full credit.

- (a) (1 point) Write the expression of the Distance IoU loss.
- (b) (1 point) Explain two advantages of the Distance IoU loss over the standard IoU loss.
- (c) (1 point) Explain one disadvantage of the Distance IoU loss over the Complete IoU loss.
- (d) (1 point) Write the expression for the binary cross-entropy loss.

Solution:

- (a) (1 point) The Distance-IoU (DIOU) loss adds a normalized distance value to the standard IoU loss as given below. {See <https://arxiv.org/pdf/1911.08287> for a detailed treatment of variants of IoU based losses.}

$$\mathcal{L}_{DIOU} = 1 - IoU + \frac{d^2(\mathbf{c}_{gt}, \mathbf{c}_{pred})}{C^2}$$

Here $d(\mathbf{c}_{gt}, \mathbf{c}_{pred})$ is the Euclidean distance between the center of the ground-truth and predicted bounding boxes. The variable C is the diagonal of the smallest rectangle containing both the predicted and the ground-truth bounding boxes.

- (b) (1 point) The distance IoU loss improved upon the standard IoU loss with the following advantages:
1. The DIOU loss does not saturate when the boxes are not overlapping, unlike the standard IoU loss.
 2. The DIOU loss is invariant to the scale of the object (since it normalizes the distances between the centers).
 3. When the ground-truth and predicted boxes have an inclusion relationship, the standard IoU loss is constant, but the DIOU loss is not. Consequently, the localization (regression) loss still generates non-zero gradients and results in improved alignment of the predicted and ground-truth bounding boxes.

- (c) (1 point) The DIoU loss does not account for discrepancies in the aspect ratio of the ground-truth and the predicted bounding boxes, which the CIoU loss does.
- (d) (1 point) For binary labels $y \in \{0, 1\}$ and predicted probability $\hat{y}$ of class 1, the binary cross-entropy loss over N samples is

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right].$$

(A sum without the $\frac{1}{N}$ is also acceptable.)